# MAT 322 HW03

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**Problem 1.** Let $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ be defined by the equation

$$f(x,y) = (x^2 - y^2, 2xy).$$

(a) Show that $f$ is injective on the set $U$ consisting of all $(x,y)$ with $x > 0$.

(b) What is the set $V = f(U)$?

(c) If $g$ is the inverse function, find $(\mathrm{d}g)_{(0,1)}$.

*Solution.* (a) Observe that we can write $f$ as the sequence of maps

$$(x,y) \mapsto x + yi \mapsto (x+yi)^2 = (x^2 - y^2) + i(2xy) \mapsto (x^2 - y^2, 2xy).$$

The first and last two maps, which swap between representing a point as a pair of coordinates and as a complex number, are bijective maps $U \to U$: we can clearly see that $(u,v) \mapsto u + vi$ is invertible with $u + vi \mapsto (u,v)$ as its inverse. The map in the middle, $z \mapsto z^2$, is injective on $U$ because for any $a \in \mathbb{C}$, $z^2 = a$ has exactly two solutions $z_1, z_2 \in \mathbb{C}$ that are negations of one another, so at most one of them has a positive real part. Compositions of injective maps are injective, so $f$ is injective on $U$.

(b) $V$ is the set

$$\mathbb{R}^2 \setminus \{(r,0) \in \mathbb{R}^2 : r \leq 0\}$$

Indeed, for $a \in \mathbb{C}$, the only time that $z^2 = a$ does not have a solution in $\mathbb{C}$ with positive real part is when both solutions have a real part of zero, which happens if and only if $a$ is a nonpositive real number.

(c) Letting $I$ be the $2 \times 2$ identity matrix, we have

$$I = (\mathrm{d}(f \circ g))_{(0,1)} = (\mathrm{d}f)_{(g(0,1))} \circ (\mathrm{d}g)_{(0,1)} = (\mathrm{d}f)_{(1,1)} \circ (\mathrm{d}g)_{(0,1)},$$

so $(\mathrm{d}g)_{(0,1)} = [(\mathrm{d}f)_{(1,1)}]^{-1}$. The derivative of $f$ in general is the matrix

$$\begin{pmatrix} 2x & 2y \\ -2y & 2x \end{pmatrix},$$

so the answer is

$$(\mathrm{d}g)_{(0,1)} = [(\mathrm{d}f)_{(1,1)}]^{-1} = \begin{pmatrix} 2 & 2 \\ -2 & 2 \end{pmatrix}^{-1} = \boxed{\begin{pmatrix} 1/4 & 1/4 \\ -1/4 & 1/4 \end{pmatrix}}.$$

□

**Problem 2.** Let $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ be defined by the equation

$$f(x,y) = (e^x \cos y, e^x \sin y).$$

(a) Show that $f$ is injective on the set $U$ consisting of all $(x,y)$ with $0 < y < 2\pi$.

(b) What is the set $V = f(U)$?

(c) If $g$ is the inverse function, find $(\mathrm{d}g)_{(0,1)}$.

*Solution.* (a) Suppose that $f(x,y) = f(x',y')$, so that

$$(e^x \cos y, e^x \sin y) = (e^{x'} \cos y', e^{x'} \sin y').$$

Then the magnitudes of these vectors are the same, so $e^x = e^{x'}$, which implies $x = x'$ because exponential functions are injective on $\mathbb{R}$. This in turn implies that $(\cos y, \sin y) = (\cos y', \sin y')$, which tells us that $y = y'$ in light of the restriction on the size of $y$.

(b) $V$ is the set $\mathbb{R}^2 \smallsetminus \{(t,0) : t \geq 0\}$. Indeed, for any point $(r\cos\theta, r\sin\theta)$ with $r > 0$ and $\theta \neq 0$ (hence in $V$), we can set $x = \ln r$, $y = \theta$ to find a point whose image under $f$ is in $V$. Conversely, if $(a,b)$ is not in $V$, then we either get $(a,b) = (0,0)$, so $e^x = 0$, which has no real solutions, or $(a,b) = (t,0)$ for some $t > 0$, giving

$$\cos y = 1, \quad \sin y = 0$$

so $y = 2\pi n$ for some $n \in \mathbb{Z}$, which is not in the interval $(0, 2\pi)$.

(c) Noting that $g(0,1) = (0, \frac{\pi}{2})$, we get

$$(\mathrm{d}g)_{(0,1)} = [(\mathrm{d}f)_{(0,\frac{\pi}{2})}]^{-1} = \begin{pmatrix} e^0 \cos \pi/2 & e^0 \sin \pi/2 \\ -e^0 \sin \pi/2 & e^0 \cos \pi/2 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{-1} = \boxed{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}}. \qquad \square$$

**Problem 3.** Let $f\colon \mathbb{R}^n \to \mathbb{R}^n$ be given by the equation $f(x) = \|x\|^2 \cdot x$. Show that $f$ is continuously differentiable and that $f$ carries the open ball $B_1(0)$ onto itself in an injective fashion. Show, however, that the inverse of $f$ is not differentiable at 0.

*Solution.* Letting $x = (x_1, \ldots, x_n)$, we can write down

$$f(x) = \begin{pmatrix} x_1(x_1^2 + x_2^2 + \cdots + x_n^2) \\ x_2(x_1^2 + x_2^2 + \cdots + x_n^2) \\ \vdots \\ x_n(x_1^2 + x_2^2 + \cdots + x_n^2) \end{pmatrix}.$$

Then we can compute the Jacobian matrix

$$(\mathrm{d}f)_x = \begin{pmatrix} 3x_1^2 & 2x_1x_2 & \cdots & 2x_1x_n \\ 2x_2x_1 & 3x_2^2 & \cdots & 2x_2x_n \\ \vdots & \vdots & \ddots & \vdots \\ 2x_nx_1 & 2x_nx_2 & \cdots & 3x_n^2 \end{pmatrix}$$

whose entries are clearly continuous in $x_1, \ldots, x_n$, so $f$ is continuously differentiable.

To show that $f$ is a bijection on $B_1(0)$, I claim that the function $g(x) = \frac{x}{\|x\|^{2/3}}$ is the inverse of $f$ on this ball. Indeed,

$$g(f(x)) = g(\|x\|^2 \cdot x) = \frac{\|x\|^2 \cdot x}{\|\|x\|^2 \cdot x\|^{2/3}} = x$$

$$f(g(x)) = f\left(\frac{x}{\|x\|^{2/3}}\right) = \left\|\frac{x}{\|x\|^{2/3}}\right\|^2 \cdot \frac{x}{\|x\|^{2/3}} = x.$$

This inverse is not differentiable at 0 because its partials are not defined at zero. For example, the limit

$$\lim_{h\to 0} \frac{f(h,0,\ldots,0) - f(0,0,\ldots,0)}{h} = \lim_{h \to 0} (1/h^{2/3}, 0, \ldots, 0)$$

does not exist. $\square$

**Problem 4.** Let $g\colon \mathbb{R}^2 \to \mathbb{R}^2$ be given by the equation

$$g(x,y) = (2ye^{2x}, xe^y)$$

and let $f\colon \mathbb{R}^2 \to \mathbb{R}^3$ be given by the equation

$$f(x,y) = (3x - y^2, 2x + y, xy + y^3).$$

(a) Show that there is a neighborhood of $(0,1)$ that $g$ carries injectively onto a neighborhood of $(2,0)$.

(b) Find $(\mathrm{d}(f \circ g^{-1}))_{(2,0)}$.

*Solution.* Observe that $g$ is a class $C^1$ function and that

$$(\mathrm{d}g)_{(0,1)} = \begin{pmatrix} 4 & 2 \\ e & 0 \end{pmatrix}$$

is invertible, so the inverse function theorem says that there is a neighborhood $U$ of $(0,1)$ that $g$ carries bijectively to a neighborhood of $g(0,1) = (2,0)$ (which is part (a)) and that $g$ has a continuously differentiable inverse on the set $g(U)$.

For part (b) we can compute

$$\begin{aligned}
(\mathrm{d}(f \circ g^{-1}))_{(2,0)} &= (\mathrm{d}f)_{g^{-1}(2,0)} \circ (\mathrm{d}g^{-1})_{(2,0)} \\
&= (\mathrm{d}f)_{g^{-1}(2,0)} \circ [(\mathrm{d}g)_{g^{-1}(2,0)}]^{-1} \\
&= (\mathrm{d}f)_{(0,1)} \circ [(\mathrm{d}g)_{(0,1)}]^{-1} \\
&= \begin{pmatrix} 3 & -2 \\ 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1/e \\ 1/2 & -2/e \end{pmatrix} \\
&= \begin{pmatrix} -1 & 7/e \\ 1/2 & 0 \\ 1 & -3/e \end{pmatrix}.
\end{aligned}$$

□

**Problem 5.** Let $U \subseteq \mathbb{R}^n$ be open and let $f\colon U \to \mathbb{R}^n$ be continuously differentiable. If $(\mathrm{d}f)_x$ is invertible for all $x \in U$, prove that $f(U)$ is an open set.

*Solution.* Let $x$ be a point in $U$. Since $f$ is a $C^1$ function with invertible derivative, we can see from the proof of the inverse function theorem that there is an open ball $B_x \subseteq U$ such that $f(B_x) \subseteq f(U)$ is open. This means that we can write

$$f(U) = \bigcup_{x \in U} f(B_x)$$

as a union of open sets, hence $f(U)$ is also open. □